

Module 4

The Time Value of Money

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Goals

- **Part 1**
 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)

Goals

- **Part 1**
 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)
- **Part 2**
 - Financial calculators

Goals

- **Part 1**

- Translate dollars today into dollars tomorrow (“future value”)
- Translate dollars tomorrow into dollars today (“present value”)

- **Part 2**

- Financial calculators

- **Part 3**

- Special cases: perpetuities, annuities, and their variants

Goals

- **Part 1**
 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)
- **Part 2**
 - Financial calculators
- **Part 3**
 - Special cases: perpetuities, annuities, and their variants
- **Part 4**
 - The role of inflation

Module 4, Part 1

Future Value and Present Value

The Time Value of Money



Would you rather receive a dollar today, or a dollar tomorrow?

The Time Value of Money

- A dollar today is worth more than a dollar tomorrow



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!
 - Assuming, there is some positive rate of return investment for us to pursue

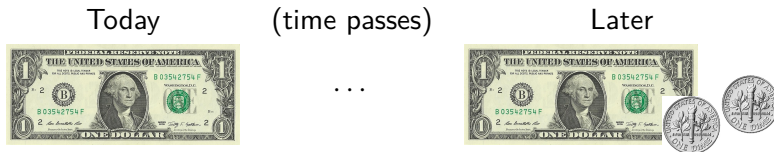


Future Values

- **Future value**, the amount to which an investment will grow after earning interest
- A “present value” today grows to its “future value” tomorrow via interest

Future Values

- **Future value**, the amount to which an investment will grow after earning interest
- A “present value” today grows to its “future value” tomorrow via interest



Future Value

- What is the future value (FV) in year T of an investment today (with an amount of PV) that earns an annual interest rate of r ?

Future Value

- What is the future value (FV) in year T of an investment today (with an amount of PV) that earns an annual interest rate of r ?

$$\text{Future value (FV)} = \text{PV} \times \underbrace{(1+r) \times (1+r) \times \cdots \times (1+r)}_{\times (1+r) \text{ repeats for } T \text{ times}}$$

Future Value

- What is the future value (FV) in year T of an investment today (with an amount of PV) that earns an annual interest rate of r ?

$$\text{Future value (FV)} = \text{PV} \times \underbrace{(1+r) \times (1+r) \times \cdots \times (1+r)}_{\times (1+r) \text{ repeats for } T \text{ times}}$$

$$\text{Future value (FV)} = \text{PV} \times (1+r)^T$$

Example, Future Value

- What is the future value (FV) of \$100 that earns an annual interest rate of 6% for 5 years?

Example, Future Value

- What is the future value (FV) of \$100 that earns an annual interest rate of 6% for 5 years?

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

Example, Future Value

- What is the future value (FV) of \$100 that earns an annual interest rate of 6% for 5 years?

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

Compound Interest

- **Compound interest**, the interest you earn with compounding (with interest on interest)

Compound Interest

- **Compound interest**, the interest you earn with compounding (with interest on interest)
 - If you earn r per year for T years, your money grows to $PV \times (1 + r)^T$
 - PV of that total is the principal, so the total compound interest you have earned is $PV \times ((1 + r)^T - 1)$

Compound Interest

- **Compound interest**, the interest you earn with compounding (with interest on interest)
 - If you earn r per year for T years, your money grows to $PV \times (1 + r)^T$
 - PV of that total is the principal, so the total compound interest you have earned is $PV \times ((1 + r)^T - 1)$
- Compound interest is the default in finance (e.g., mortgage rate, deposit rate)

Future Value, Compound Interest

- You have \$100 in a bank account, and your bank pays 6% per year in interest
- How much interest will you have earned after five years?

Future Value, Compound Interest

- You have \$100 in a bank account, and your bank pays 6% per year in interest
- How much interest will you have earned after five years?

Year	Start balance	Compound interest	End balance
1	\$100.00	$0.06 \times 100.00 = \$6.00$	\$106.00
2	\$106.00	$0.06 \times 106.00 = \$6.36$	\$112.36
3	\$112.36	$0.06 \times 112.36 = \$6.74$	\$119.10
4	\$119.10	$0.06 \times 119.10 = \$7.15$	\$126.25
5	\$126.25	$0.06 \times 126.25 = \$7.57$	\$133.82

Future Value, Compound Interest

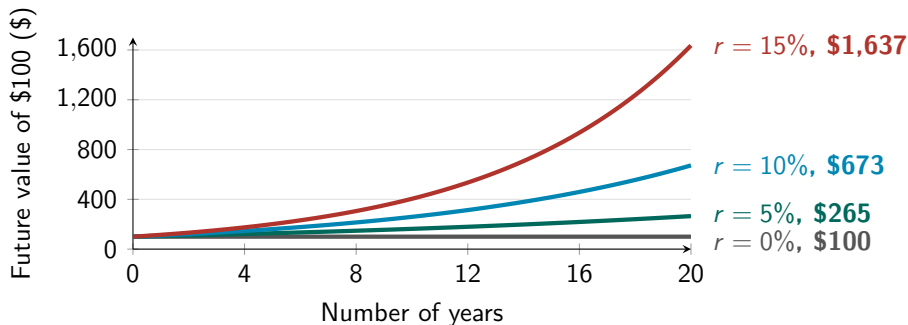
- You have \$100 in a bank account, and your bank pays 6% per year in interest
- How much interest will you have earned after five years?

Year	Start balance	Compound interest	End balance
1	\$100.00	$0.06 \times 100.00 = \$6.00$	\$106.00
2	\$106.00	$0.06 \times 106.00 = \$6.36$	\$112.36
3	\$112.36	$0.06 \times 112.36 = \$6.74$	\$119.10
4	\$119.10	$0.06 \times 119.10 = \$7.15$	\$126.25
5	\$126.25	$0.06 \times 126.25 = \$7.57$	\$133.82

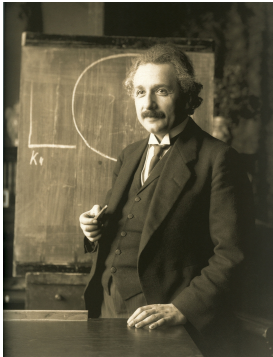
- With compound interest, your \$100 grows to $\$100(1 + r)^T = 133.82$, so you earned \$33.82 in compound interest

Future Value

- The future value (FV) depends heavily on the interest rate r , especially as the time dimension T grows

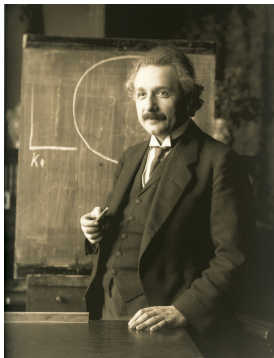


What's the Most Powerful Force in the Universe?



- Einstein, ...

What's the Most Powerful Force in the Universe?



- Einstein, ...
- “Compound interest”

Returns, Holding Period and Annualized

- **Return**, percentage growth of an asset's value over some period of time
 - E.g., if you earn an interest of 5% over a year, then your return will be 5%

Returns, Holding Period and Annualized

- **Return**, percentage growth of an asset's value over some period of time
 - E.g., if you earn an interest of 5% over a year, then your return will be 5%
- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for T periods with returns $r_1, r_2, \dots, r_T$ gives $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \dots (1 + r_T)$

Returns, Holding Period and Annualized

- **Return**, percentage growth of an asset's value over some period of time
 - E.g., if you earn an interest of 5% over a year, then your return will be 5%
- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for T periods with returns $r_1, r_2, \dots, r_T$ gives $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \dots (1 + r_T)$
- **Annualized return**, percentage growth of money per year, calculated as a geometric mean over the T years of the holding period

$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1$$

Returns, Holding Period and Annualized

- **Return**, percentage growth of an asset's value over some period of time
 - E.g., if you earn an interest of 5% over a year, then your return will be 5%
- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for T periods with returns $r_1, r_2, \dots, r_T$ gives $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \dots (1 + r_T)$
- **Annualized return**, percentage growth of money per year, calculated as a geometric mean over the T years of the holding period

$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1$$

- These financial returns are not directly related in any way with **Module 3's** accounting-based ROA and ROE measures

Example, Annualized Returns

- Peloton had a return of 434% in 2020, and -76% in 2021
- What was the (compound) holding period return for holding Peloton's stock over these two years?
- What was the annualized return?

Example, Annualized Returns

- Peloton had a return of 434% in 2020, and -76% in 2021
- What was the (compound) holding period return for holding Peloton's stock over these two years?
- What was the annualized return?

$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 5.34 \times 0.24 - 1 = 28.2\%$$

Example, Annualized Returns

- Peloton had a return of 434% in 2020, and -76% in 2021
- What was the (compound) holding period return for holding Peloton's stock over these two years?
- What was the annualized return?

$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 5.34 \times 0.24 - 1 = 28.2\%$$

$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1 = 1.282^{0.5} - 1 = 13.2\%$$

Example, Annualized Returns

- Peloton had a return of 434% in 2020, and -76% in 2021
- What was the (compound) holding period return for holding Peloton's stock over these two years?
- What was the annualized return?

$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 5.34 \times 0.24 - 1 = 28.2\%$$

$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1 = 1.282^{0.5} - 1 = 13.2\%$$

- Notice that the annual return is NOT $28.2\%/2 = 14.1\%$

Example, Annualized Returns

- Peloton had a return of 434% in 2020, and -76% in 2021
- What was the (compound) holding period return for holding Peloton's stock over these two years?
- What was the annualized return?

$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 5.34 \times 0.24 - 1 = 28.2\%$$

$$r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1 = 1.282^{0.5} - 1 = 13.2\%$$

- Notice that the annual return is NOT $28.2\%/2 = 14.1\%$
- Confirm, you can also verify that $(1 + 0.132)^2 - 1 = 28.2\%$

Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?



Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?
- a. $FV = \$24 \times (1 + 0.08)^{395} = \382 trillion (roughly the same as total world wealth)



Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?
- a. $FV = \$24 \times (1 + 0.08)^{395} = \382 trillion (roughly the same as total world wealth)
- b. $FV = \$24 \times (1 + 0.035)^{395} = \19 million



Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?



Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?
- 11.896 years, because $(1 + 0.06)^{11.896} = 2$



Present Value

- **Present value**, the value today of a future cash flow

Present Value

- **Present value**, the value today of a future cash flow
- Recall, Future value $(FV) = PV \times (1 + r)^T$, that means

$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T} \quad (\text{the fraction is the "discount factor"})$$

Present Value

- **Present value**, the value today of a future cash flow
- Recall, Future value $(FV) = PV \times (1 + r)^T$, that means

$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T} \quad (\text{the fraction is the "discount factor"})$$

- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value

Present Value

- **Present value**, the value today of a future cash flow
- Recall, Future value $(FV) = PV \times (1 + r)^T$, that means

$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T} \quad (\text{the fraction is the "discount factor"})$$

- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value
- **Discount rate**, the rate used to compute PVs (i.e., r)
 - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
 - For a (systematically) risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in **Modules 8 and 9**

Present Value

- **Present value**, the value today of a future cash flow
- Recall, Future value (FV) = $PV \times (1 + r)^T$, that means

$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T} \quad (\text{the fraction is the "discount factor"})$$

- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value
- **Discount rate**, the rate used to compute PVs (i.e., r)
 - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
 - For a (systematically) risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in [Modules 8 and 9](#)
- **Discount factor**, the present value of \$1 received in year T when the discount rate is r , less than one when r is positive

Present Value

- **Present value**, the value today of a future cash flow
- Recall, Future value $(FV) = PV \times (1 + r)^T$, that means

$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T} \quad (\text{the fraction is the "discount factor"})$$

- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value
- **Discount rate**, the rate used to compute PVs (i.e., r)
 - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
 - For a (systematically) risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in **Modules 8 and 9**
- **Discount factor**, the present value of \$1 received in year T when the discount rate is r , less than one when r is positive
- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

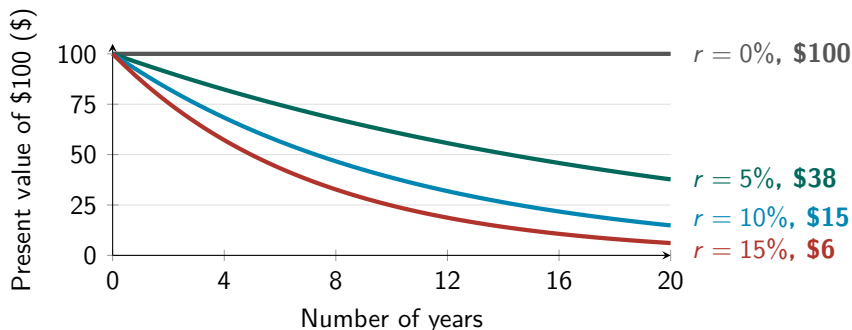
Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

$$\text{Present value (PV)} = \frac{\text{FV}}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

Present Value

- The present value (PV) of \$100 in the future depends heavily on the discount rate r , especially as the time dimension T grows



Present Value, Multiple Future Cash Flows

- Present values can be added together to calculate the total PV of multiple future cash flows

Present Value, Multiple Future Cash Flows

- Present values can be added together to calculate the total PV of multiple future cash flows

$$PV_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots$$

- where C_1 is the cash flow in one year, C_2 is the cash flow in two years, etc.

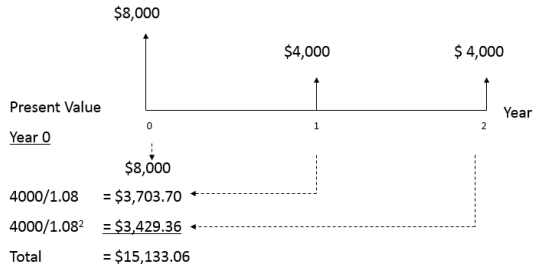
Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

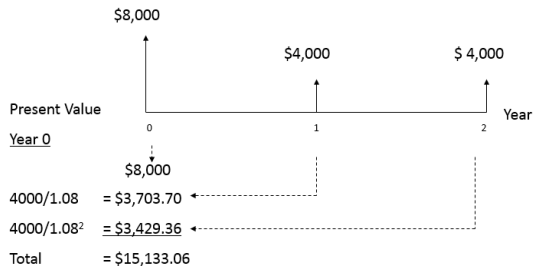
$$PV = \$8,000 + \frac{\$4,000}{(1 + 0.08)^1} + \frac{\$4,000}{(1 + 0.08)^2}$$



Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

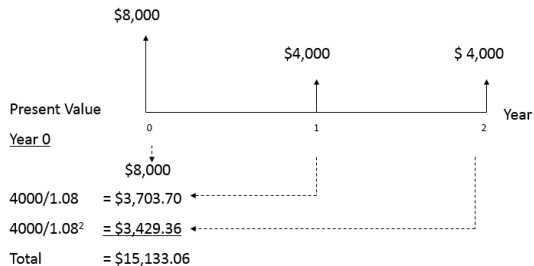
$$\begin{aligned}PV &= \$8,000 + \frac{\$4,000}{(1 + 0.08)^1} + \frac{\$4,000}{(1 + 0.08)^2} \\&= \$15,133.06\end{aligned}$$



Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

$$\begin{aligned}PV &= \$8,000 + \frac{\$4,000}{(1 + 0.08)^1} + \frac{\$4,000}{(1 + 0.08)^2} \\ &= \$15,133.06\end{aligned}$$

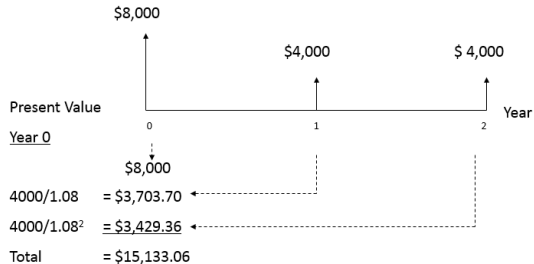


- The present value (\$15,133.06) is less than \$15,500, so choose the payment plan

Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

$$\begin{aligned} PV &= \$8,000 + \frac{\$4,000}{(1 + 0.08)^1} + \frac{\$4,000}{(1 + 0.08)^2} \\ &= \$15,133.06 \end{aligned}$$



- The present value (\$15,133.06) is less than \$15,500, so choose the payment plan
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

- The present value of a future cash flow is the value of that cash flow multiplied by its discount factor

$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1 + r)^T}$$

Recap of Part 1

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest

$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

- The present value of a future cash flow is the value of that cash flow multiplied by its discount factor

$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1 + r)^T}$$

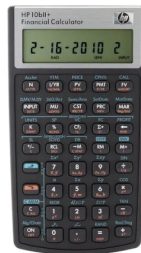
- Present values can be added together to calculate the total present value of multiple future cash flows

Module 4, Part 2

Financial Calculators

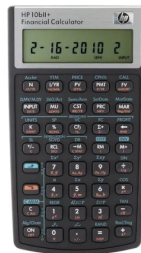
Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)



Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)
- A complement to the underlying math and intuition, never a substitute



Reset Your Financial Calculator

- If your financial calculator yields wrong answers, you may have changed some settings accidentally!

Reset Your Financial Calculator

- If your financial calculator yields wrong answers, you may have changed some settings accidentally!
- To reset your HP 10bII+, press and hold the ON, N, and PV keys simultaneously. Release all three keys at once, and the display will show “ALL CLEAR” to confirm the reset

Reset Your Financial Calculator

- If your financial calculator yields wrong answers, you may have changed some settings accidentally!
- To reset your HP 10bII+, press and hold the ON, N, and PV keys simultaneously. Release all three keys at once, and the display will show “ALL CLEAR” to confirm the reset
- To reset your TI BA II Plus, press the 2nd key, then the $\div$ key (which has “RESET” written above it), and finally the ENTER key to confirm the “RST?” prompt

Reset Your Financial Calculator

- If your financial calculator yields wrong answers, you may have changed some settings accidentally!
- To reset your HP 10bII+, press and hold the ON, N, and PV keys simultaneously. Release all three keys at once, and the display will show “ALL CLEAR” to confirm the reset
- To reset your TI BA II Plus, press the 2nd key, then the $\div$ key (which has “RESET” written above it), and finally the ENTER key to confirm the “RST?” prompt
- After you reset, you should repeat the next two settings!

Set the Number of Significant Digits to at Least 4!

- On the HP 10bII+, to change the number of digits displayed to 4 decimals, press
 - OrangeOption
 - = (DISP)
 - 4

Set the Number of Significant Digits to at Least 4!

- On the HP 10bII+, to change the number of digits displayed to 4 decimals, press
 - OrangeOption
 - = (DISP)
 - 4
- On the TI BA II Plus, press
 - 2nd
 - .
 - 4
 - ENTER

Set the Number of Compounding Periods per Year to 1!

- On the HP 10bII+, to set it to 1,
press
 - 1
 - OrangeOption
 - PMT (P/YR)

Set the Number of Compounding Periods per Year to 1!

- On the HP 10bII+, to set it to 1, press
 - 1
 - OrangeOption
 - PMT (P/YR)
- To change back to the factory default (12), press
 - 12
 - OrangeOption
 - PMT (P/YR)

Set the Number of Compounding Periods per Year to 1!

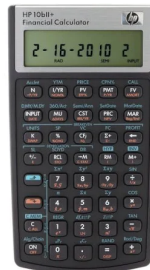
- On the HP 10bII+, to set it to 1, press
 - 1
 - OrangeOption
 - PMT (P/YR)
- To change back to the factory default (12), press
 - 12
 - OrangeOption
 - PMT (P/YR)
- On the TI BA II Plus, press
 - 2nd
 - I/Y (P/Y)
 - 1
 - ENTER

Set the Number of Compounding Periods per Year to 1!

- On the HP 10bII+, to set it to 1, press
 - 1
 - OrangeOption
 - PMT (P/YR)
- To change back to the factory default (12), press
 - 12
 - OrangeOption
 - PMT (P/YR)
- On the TI BA II Plus, press
 - 2nd
 - I/Y (P/Y)
 - 1
 - ENTER
- You must do this!

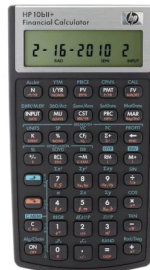
Financial Calculators, Basic Functionality

- N, number of periods (usually years)



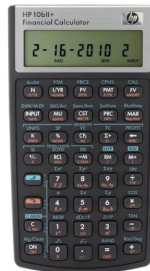
Financial Calculators, Basic Functionality

- N , number of periods (usually years)
- I/YR , interest rate, expressed as a percentage (not a decimal)



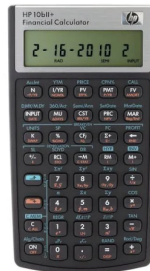
Financial Calculators, Basic Functionality

- N, number of periods (usually years)
- I/YR, interest rate, expressed as a percentage (not a decimal)
- PV, present value



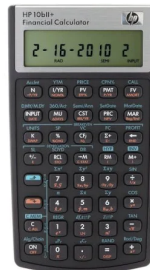
Financial Calculators, Basic Functionality

- N, number of periods (usually years)
- I/YR, interest rate, expressed as a percentage (not a decimal)
- PV, present value
- PMT, recurring payment amount



Financial Calculators, Basic Functionality

- N, number of periods (usually years)
- I/YR, interest rate, expressed as a percentage (not a decimal)
- PV, present value
- PMT, recurring payment amount
- FV, future value (lump sum)



Financial Calculators, Basic Functionality

- Built for problems of the type, the PV of N periodic payments of PMT plus a terminal lump sum FV, discounted at the rate I , i.e., $PV = f(N, PMT, FV, I)$



Financial Calculators, Basic Functionality

- Built for problems of the type, the PV of N periodic payments of PMT plus a terminal lump sum FV, discounted at the rate I , i.e., $PV = f(N, PMT, FV, I)$



- But also, any of the five from the other four,
 - $N = f(PV, PMT, FV, I)$

Financial Calculators, Basic Functionality

- Built for problems of the type, the PV of N periodic payments of PMT plus a terminal lump sum FV, discounted at the rate I , i.e., $PV = f(N, PMT, FV, I)$



- But also, any of the five from the other four,
 - $N = f(PV, PMT, FV, I)$
 - $PMT = f(PV, N, FV, I)$

Financial Calculators, Basic Functionality

- Built for problems of the type, the PV of N periodic payments of PMT plus a terminal lump sum FV, discounted at the rate I , i.e., $PV = f(N, PMT, FV, I)$



- But also, any of the five from the other four,
 - $N = f(PV, PMT, FV, I)$
 - $PMT = f(PV, N, FV, I)$
 - $I = f(PV, PMT, FV, N)$

Financial Calculators, Basic Functionality

- Built for problems of the type, the PV of N periodic payments of PMT plus a terminal lump sum FV, discounted at the rate I , i.e., $PV = f(N, PMT, FV, I)$



- But also, any of the five from the other four,
 - $N = f(PV, PMT, FV, I)$
 - $PMT = f(PV, N, FV, I)$
 - $I = f(PV, PMT, FV, N)$
 - $FV = f(PV, PMT, N, I)$

Example, Using Financial Calculators (HP 10bII+)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

Example, Using Financial Calculators (HP 10bII+)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
- Enter the number, then push the financial function key,
 - 1 N
 - 8 I/YR
 - 0 PMT
 - 3000 FV
 - PV

n	i	PV	PMT	FV
2	8	PV	0	3000

Example, Using Financial Calculators (HP 10bII+)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
- Enter the number, then push the financial function key,
 - 1 N
 - 8 I/YR
 - 0 PMT
 - 3000 FV
 - PV
- The order of the first four steps does not matter!

n	i	PV	PMT	FV
2	8	PV	0	3000

Example, Using Financial Calculators (HP 10bII+)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
 - Enter the number, then push the financial function key,
 - 1 N
 - 8 I/YR
 - 0 PMT
 - 3000 FV
 - PV
 - The order of the first four steps does not matter!
- Pushing PV gives the output
—\$2,778

n	i	PV	PMT	FV
2	8	PV	0	3000

Example, Using Financial Calculators (HP 10bII+)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
- Enter the number, then push the financial function key,
 - 1 N
 - 8 I/YR
 - 0 PMT
 - 3000 FV
 - PV
- The order of the first four steps does not matter!

n	i	PV	PMT	FV
2	8	PV	0	3000

- Pushing PV gives the output
—\$2,778
- Why does the answer come out negative?

Example, Using Financial Calculators (TI BA II Plus)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

Example, Using Financial Calculators (TI BA II Plus)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
- To clear saved calculations (useful before starting a time value of money problem), press 2nd, then FV (ClrTVM)

Example, Using Financial Calculators (TI BA II Plus)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
- To clear saved calculations (useful before starting a time value of money problem), press 2nd, then FV (ClrTVM)
- Solution,
 - 1 N
 - 8 I/YR
 - 0 PMT
 - 3000 FV
 - CPT (“compute”), then PV

Example, Using Financial Calculators (TI BA II Plus)

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?
- To clear saved calculations (useful before starting a time value of money problem), press 2nd, then FV (ClrTVM)
- Solution,
 - 1 N
 - 8 I/YR
 - 0 PMT
 - 3000 FV
 - CPT (“compute”), then PV
- The order of the first four steps does not matter, and the output is $-\$2,778$

The Equation the Calculator Solves

- We solved this problem earlier in this module using the following formula

$$\text{Present value} = PV = \frac{FV}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

The Equation the Calculator Solves

- We solved this problem earlier in this module using the following formula

$$\text{Present value} = PV = \frac{FV}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

- But your financial calculator is instead programmed to solve this equation

$$PV + \left[\frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

The Equation the Calculator Solves

- We solved this problem earlier in this module using the following formula

$$\text{Present value} = PV = \frac{FV}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

- But your financial calculator is instead programmed to solve this equation

$$PV + \left[\frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

- Note, here all the $PMT = 0$ since there is only a lump sum at the end

The Equation the Calculator Solves

- We solved this problem earlier in this module using the following formula

$$\text{Present value} = PV = \frac{FV}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

- But your financial calculator is instead programmed to solve this equation

$$PV + \left[\frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

- Note, here all the $PMT = 0$ since there is only a lump sum at the end
- So the PV will have the opposite sign of the payments (PMT) and the future value (FV)

Example, Puerto Rico Borrows

- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
- It did so by selling bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each (they sold around $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$ bonds, and thus promised to pay back around \$27.5 billion in total)
- What was the interest rate (i) paid by Puerto Rico?



Example, Puerto Rico Borrows

- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
- It did so by selling bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each (they sold around $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$ bonds, and thus promised to pay back around \$27.5 billion in total)
- What was the interest rate (I) paid by Puerto Rico?



- Solution,
 - 47 N
 - -94.40 PV
 - 0 PMT
 - 1000 FV
 - I/YR

Example, Puerto Rico Borrows

- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
- It did so by selling bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each (they sold around $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$ bonds, and thus promised to pay back around \$27.5 billion in total)
- What was the interest rate (I) paid by Puerto Rico?



- Solution,
 - 47 N
 - -94.40 PV
 - 0 PMT
 - 1000 FV
 - I/YR
- The rate is 5.15%

Example, Mortgage

- Suppose you are applying for a 360-month (30-year) mortgage, where the monthly interest rate is 0.40%. What will the monthly payments be if you want to borrow \$400,000?



Example, Mortgage

- Suppose you are applying for a 360-month (30-year) mortgage, where the monthly interest rate is 0.40%. What will the monthly payments be if you want to borrow \$400,000?
- Solution, we need to solve for PMT,
 - $-400,000$ PV
 - 360 N
 - 0.4 I/YR
 - 0 FV (no lump sum at the end)
 - PMT



Example, Mortgage

- Suppose you are applying for a 360-month (30-year) mortgage, where the monthly interest rate is 0.40%. What will the monthly payments be if you want to borrow \$400,000?
- Solution, we need to solve for PMT,
 - $-400,000$ PV
 - 360 N
 - 0.4 I/YR
 - 0 FV (no lump sum at the end)
 - PMT
- The monthly payment is \$2,098.66



Example, Mortgage

- Suppose you are applying for a 360-month (30-year) mortgage, where the monthly interest rate is 0.40%. What will the monthly payments be if you want to borrow \$400,000?
- Solution, we need to solve for PMT,
 - $-400,000$ PV
 - 360 N
 - 0.4 I/YR
 - 0 FV (no lump sum at the end)
 - PMT
- The monthly payment is \$2,098.66



- That is, you would pay an undiscounted total of $360 \times 2,098.66$ over the course of your mortgage

Example, Mortgage

- Suppose you are applying for a 360-month (30-year) mortgage, where the monthly interest rate is 0.40%. What will the monthly payments be if you want to borrow \$400,000?
- Solution, we need to solve for PMT,
 - $-400,000$ PV
 - 360 N
 - 0.4 I/YR
 - 0 FV (no lump sum at the end)
 - PMT
- The monthly payment is \$2,098.66



- That is, you would pay an undiscounted total of $360 \times 2,098.66$ over the course of your mortgage
- Bonus, what would the payment be if the monthly interest was 0.2%?

Example, Pick Your Payment!

- Many car dealerships offer “Pick Your Payment”, you can choose how much to pay (PMT) every month! What usually varies is N (the number of months payments continue)
- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?

Example, Pick Your Payment!

- Many car dealerships offer “Pick Your Payment”, you can choose how much to pay (PMT) every month! What usually varies is N (the number of months payments continue)
- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?
- Joe, solve for N ,
 - -20,000 PV
 - 0.6 I/YR
 - 0 FV
 - 400 PMT
 - N

Example, Pick Your Payment!

- Many car dealerships offer “Pick Your Payment”, you can choose how much to pay (PMT) every month! What usually varies is N (the number of months payments continue)
- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?
- Joe, solve for N ,
 - -20,000 PV
 - 0.6 I/YR
 - 0 FV
 - 400 PMT
 - N
- N is 59.6 months

Example, Pick Your Payment!

- Many car dealerships offer “Pick Your Payment”, you can choose how much to pay (PMT) every month! What usually varies is N (the number of months payments continue)
- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?
- Joe, solve for N ,
 - -20,000 PV
 - 0.6 I/YR
 - 0 FV
 - 400 PMT
 - N
- N is 59.6 months
- Lucy, solve for N ,
 - -20,000 PV
 - 0.6 I/YR
 - 0 FV
 - 600 PMT
 - N

Example, Pick Your Payment!

- Many car dealerships offer “Pick Your Payment”, you can choose how much to pay (PMT) every month! What usually varies is N (the number of months payments continue)
- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?
- Joe, solve for N ,
 - -20,000 PV
 - 0.6 I/YR
 - 0 FV
 - 400 PMT
 - N
- N is 59.6 months
- Lucy, solve for N ,
 - -20,000 PV
 - 0.6 I/YR
 - 0 FV
 - 600 PMT
 - N
- N is 37.3 months

Excel Functionality

- Excel also has built-in functionality similar to financial calculators

Excel Functionality

- Excel also has built-in functionality similar to financial calculators
- Functions,
 - `PV(rate, nper, pmt, [fv])`
 - `FV(rate, nper, pmt, [pv])`
 - `RATE(nper, pmt, pv, [fv], [type], [guess])`
 - `PMT(rate, nper, pv, [fv])`
 - `NPER(rate, pmt, pv, [fv])`

Excel Functionality

- Excel also has built-in functionality similar to financial calculators
- Functions,
 - `PV(rate, nper, pmt, [fv])`
 - `FV(rate, nper, pmt, [pv])`
 - `RATE(nper, pmt, pv, [fv], [type], [guess])`
 - `PMT(rate, nper, pv, [fv])`
 - `NPER(rate, pmt, pv, [fv])`
- Note, unlike for financial calculators, in Excel the rate is in decimals (e.g., 0.02), not percent (e.g., 2%)!

Excel Functionality

- Excel also has built-in functionality similar to financial calculators
- Functions,
 - `PV(rate, nper, pmt, [fv])`
 - `FV(rate, nper, pmt, [pv])`
 - `RATE(nper, pmt, pv, [fv], [type], [guess])`
 - `PMT(rate, nper, pv, [fv])`
 - `NPER(rate, pmt, pv, [fv])`
- Note, unlike for financial calculators, in Excel the rate is in decimals (e.g., 0.02), not percent (e.g., 2%)!
- Let's try to solve some of the previous examples using Excel!

Example, A-Rod's \$275 Million Contract with the Yankees

Year	Salary	"Signing bonus"
2008	27	2
2009	32	1
2010	32	1
2011	31	1
2012	29	1
2013	28	1
2014	25	3
2015	21	
2016	20	
2017	20	
Total	265	10

- What was A-Rod's \$275 million contract with the Yankees actually worth?



Example, A-Rod's \$275 Million Contract with the Yankees

Year	Salary	"Signing bonus"
2008	27	2
2009	32	1
2010	32	1
2011	31	1
2012	29	1
2013	28	1
2014	25	3
2015	21	
2016	20	
2017	20	
Total	265	10

- What was A-Rod's \$275 million contract with the Yankees actually worth?
- Suppose the discount rate is 10%. What is the present value of his contract?



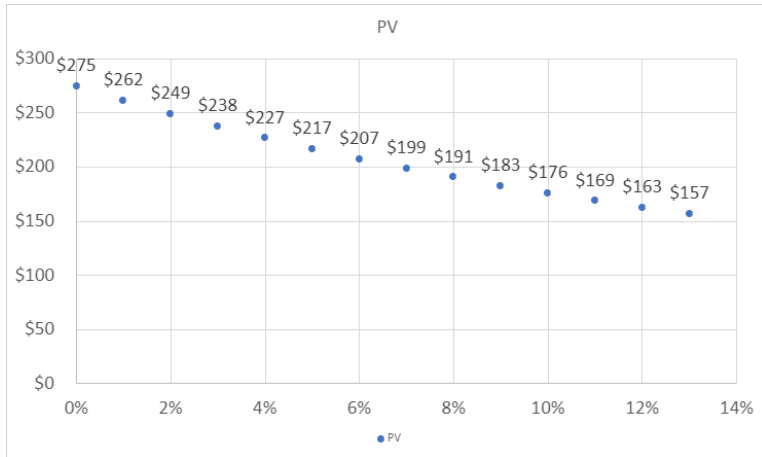
Example, A-Rod's \$275 Million Contract with the Yankees

Year	Salary	"Signing bonus"
2008	27	2
2009	32	1
2010	32	1
2011	31	1
2012	29	1
2013	28	1
2014	25	3
2015	21	
2016	20	
2017	20	
Total	265	10

- What was A-Rod's \$275 million contract with the Yankees actually worth?
- Suppose the discount rate is 10%. What is the present value of his contract?
- Why might it be useful for A-Rod to know the PV, and not simply the (undiscounted) sum of all future payments?



The PV of Payments to A-Rod



Recap of Part 2

- Financial calculators solve any one of N , I/YR , PV , PMT , and FV from the other four

Recap of Part 2

- Financial calculators solve any one of N , I/YR , PV , PMT , and FV from the other four
- The PV has the opposite sign of the payments (PMT) and the future value (FV)

Recap of Part 2

- Financial calculators solve any one of N , I/YR , PV , PMT , and FV from the other four
- The PV has the opposite sign of the payments (PMT) and the future value (FV)
- Set the number of compounding periods per year to 1, and show at least 4 significant digits

Recap of Part 2

- Financial calculators solve any one of N , I/YR , PV , PMT , and FV from the other four
- The PV has the opposite sign of the payments (PMT) and the future value (FV)
- Set the number of compounding periods per year to 1, and show at least 4 significant digits
- Excel has the same functionality, with the rate in decimals

Module 4, Part 3

Special Cases: Perpetuities and Annuities

Perpetuities and Annuities

- **Perpetuity**, a level stream of cash flows at regular intervals (the same amount every period), that never ends

Perpetuities and Annuities

- **Perpetuity**, a level stream of cash flows at regular intervals (the same amount every period), that never ends
- **Annuity**, a level stream of cash flows at regular intervals with a finite maturity (these cash flows end after N periods)

Perpetuities and Annuities

- **Perpetuity**, a level stream of cash flows at regular intervals (the same amount every period), that never ends
- **Annuity**, a level stream of cash flows at regular intervals with a finite maturity (these cash flows end after N periods)
- Why care about perpetuities and annuities?

Perpetuities and Annuities

- **Perpetuity**, a level stream of cash flows at regular intervals (the same amount every period), that never ends
- **Annuity**, a level stream of cash flows at regular intervals with a finite maturity (these cash flows end after N periods)
- Why care about perpetuities and annuities?
 - Many real-world cash flows have these patterns (particularly annuities, e.g., most mortgages or other fixed-term loans)

Perpetuities and Annuities

- **Perpetuity**, a level stream of cash flows at regular intervals (the same amount every period), that never ends
- **Annuity**, a level stream of cash flows at regular intervals with a finite maturity (these cash flows end after N periods)
- Why care about perpetuities and annuities?
 - Many real-world cash flows have these patterns (particularly annuities, e.g., most mortgages or other fixed-term loans)
 - There exist neat shortcuts that make it very easy to calculate the present value of these patterns of cash flows

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



- Notice that,
 - The first payment happens after one period (not immediately)
 - This is in some sense an “infinite” amount of money since it never ends!

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



- Notice that,
 - The first payment happens after one period (not immediately)
 - This is in some sense an “infinite” amount of money since it never ends!
- A perpetuity has a present value (PV) of

$$\text{PV of perpetuity} = \frac{\text{Cash flow}}{\text{Discount rate}} = \frac{C}{r}$$

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



- Notice that,
 - The first payment happens after one period (not immediately)
 - This is in some sense an “infinite” amount of money since it never ends!
- A perpetuity has a present value (PV) of

$$\text{PV of perpetuity} = \frac{\text{Cash flow}}{\text{Discount rate}} = \frac{C}{r}$$

- C is the annual cash flow (aka the payment), and r is the discount rate

Example, Perpetuities

- Recall A-Rod's Yankees contract. Suppose the discount rate is 10%, then the PV of the contract (paying \$275m over 10 years) is \$175.6m
- Now suppose the Yankees offer A-Rod two options,
 - A. The contract as previously described (10 years with \$275m in total), or
 - B. A perpetual payment of C dollars every year. He will play for 10 years, but continue to be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- How much should the Yankees offer as the perpetual payment C for him to be **indifferent** between contract (A) and the “being paid forever” contract (B)?



Example, Perpetuities

- Solution, he is indifferent if the contracts have the same present value!

Example, Perpetuities

- Solution, he is indifferent if the contracts have the same present value!

$$PV_{\text{Contract A}} = PV_{\text{Contract B}} = \frac{C_B}{r} \Rightarrow C_B = PV \times r = 175.6 \times 10\% = \$17.56m$$

Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



- **Ordinary perpetuity**, first payment at time 1

$$PV = \frac{C}{r}$$



Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



- **Ordinary perpetuity**, first payment at time 1

$$PV = \frac{C}{r}$$



- **Delayed perpetuity**, first payment at time $t > 1$

$$PV = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$



Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity

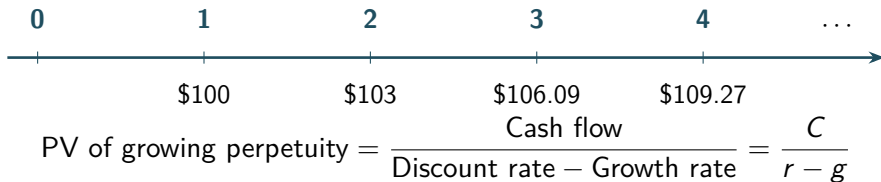
Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity
- Example, a growing perpetuity with a first payment of \$100 that grows at a rate of 3% has cash flows



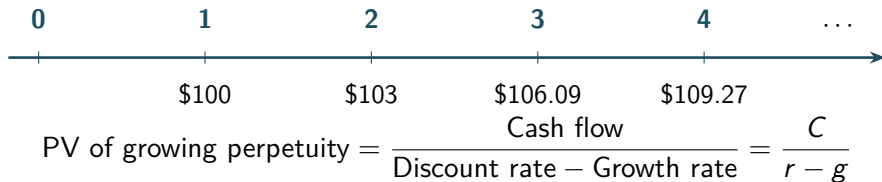
Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity
- Example, a growing perpetuity with a first payment of \$100 that grows at a rate of 3% has cash flows



Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity
- Example, a growing perpetuity with a first payment of \$100 that grows at a rate of 3% has cash flows



- This formula comes up often when valuing stocks, e.g., in the “Gordon growth model” and DCF valuation (more in [Modules 5 and 9](#))
- When $g = 0$, the PV simplifies to the formula for a regular (non-growing) perpetuity

Annuities

- An annuity is a stream of N equal cash flows paid at regular intervals



- Note, the first payment happens after one period (not immediately)

Annuities

- An annuity is a stream of N equal cash flows paid at regular intervals



- Note, the first payment happens after one period (not immediately)
- Present value of an annuity,

$$\text{PV of annuity} = C \underbrace{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)}_{\text{present value annuity factor (PVAF)}}$$

Annuities

- An annuity is a stream of N equal cash flows paid at regular intervals



- Note, the first payment happens after one period (not immediately)
- Present value of an annuity,

$$\text{PV of annuity} = C \underbrace{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)}_{\text{present value annuity factor (PVAF)}}$$

- C is the annual cash flow (aka the payment), r is the discount rate, and N is the number of years that the annuity lasts

Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

Example, Annuity

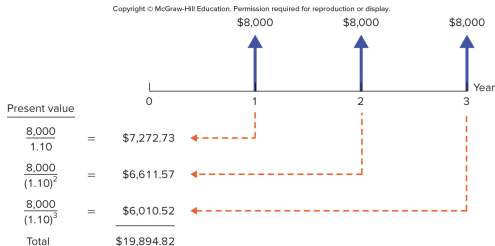
- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

$$PV = C \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left(1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$

Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

$$PV = C \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left(1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$



Example, Annuity

- Suppose the discount rate for A-Rod's contract is 10%, so the present value is \$175.6m
- Now suppose instead the Yankees offer an annuity for 10 years with the same present value
- How much would he get paid per year, and in total (undiscounted) over 10 years, under such a same-value annuity contract?



Example, Annuity

- Solution,

Example, Annuity

- Solution,

$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

Example, Annuity

- Solution,

$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

- So he would be paid \$28.58m per year, or \$285.8m in undiscounted total over 10 years (slightly more in total, because his actual contract is front-loaded, which raises its present value compared to a constant payment)

Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?

Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?
- Answer, the front-loaded one

Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?
- Answer, the front-loaded one
- What about a back-loaded cash flow and an annuity?

Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



- **Ordinary annuity**, first payment at time 1

$$PV = PV \text{ annuity}$$



Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



- **Ordinary annuity**, first payment at time 1

$$PV = PV \text{ annuity}$$



- **Delayed annuity**, first payment at time $t > 1$

$$PV = PV \text{ annuity} \times \frac{1}{(1+r)^{t-1}}$$



Practice Problem

- A store offers two payment plans. Under the installment plan, you pay 25% down and 25% of the purchase price in each of the next 3 years. If you pay the entire bill immediately, you can get a discount of 7% on the purchase price. Assume the product sells for \$100
- a-1. Calculate the present value of the payments if you can borrow or lend funds at an interest rate of 4%
- a-2. Which is a better deal?
- b-1. Calculate the present value if the payments on the 4-year installment plan do not start for a full year
- b-2. Which is a better deal?

Practice Problem

- a-1. This is an annuity due, since the first payment is due today

Practice Problem

- a-1. This is an annuity due, since the first payment is due today

This is an annuity due since the first payment is due today.

$$PV = \$25 + [\$25 \times \text{annuity factor (4\%, 3 years)}]$$

$$PV = \$25 + \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^3} \right] = \$94.38$$

Practice Problem

- a-1. This is an annuity due, since the first payment is due today

This is an annuity due since the first payment is due today.

$$PV = \$25 + [\$25 \times \text{annuity factor (4\%, 3 years)}]$$

$$PV = \$25 + \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^3} \right] = \$94.38$$

- a-2. Paying in full now is better (\$93 beats \$94.38)

Practice Problem

- a-1. This is an annuity due, since the first payment is due today

This is an annuity due since the first payment is due today.

$$PV = \$25 + [\$25 \times \text{annuity factor (4\%, 3 years)}]$$

$$PV = \$25 + \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^3} \right] = \$94.38$$

- a-2. Paying in full now is better (\$93 beats \$94.38)
- On a financial calculator, $PMT = -25$, $N = 3$, $FV = 0$, $I/YR = 4$, leading to 69.38, and next, $69.38 + 25 = 94.38$

Practice Problem

- b-1. Now it is an ordinary annuity, since the first payment is due at the end of year 1

Practice Problem

- b-1. Now it is an ordinary annuity, since the first payment is due at the end of year 1

b-1.

This is an ordinary annuity since the first payment is due at the end of Year 1.

$$PV = \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^5} \right] = \$90.75$$

Practice Problem

- b-1. Now it is an ordinary annuity, since the first payment is due at the end of year 1

b-1.

This is an ordinary annuity since the first payment is due at the end of Year 1.

$$PV = \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^5} \right] = \$90.75$$

- b-2. The installment plan is better (\$90.75 beats \$93)

Practice Problem

- b-1. Now it is an ordinary annuity, since the first payment is due at the end of year 1

b-1.

This is an ordinary annuity since the first payment is due at the end of Year 1.

$$PV = \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^4} \right] = \$90.75$$

- b-2. The installment plan is better (\$90.75 beats \$93)
- On a financial calculator, $PMT = -25$, $N = 4$, $FV = 0$, $I/YR = 4$

Practice Problem

- A local bank advertises the following deal, “Pay us \$100 at the end of each year for 11 years, and then we will pay you (or your beneficiaries) \$100 at the end of each year forever”
 - a. Calculate the present value of your payments to the bank if the interest rate is 7.00%
 - b. What is the present value of a \$100 perpetuity deferred for 11 years if the interest rate is 7.00%?
 - c. Is this a good deal?

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV
- b. The bank's payments to you are a perpetuity deferred for 11 years

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV
- b. The bank's payments to you are a perpetuity deferred for 11 years

$$PV_{11} = \frac{C}{r}$$
$$PV_{11} = \frac{100}{7\%} = \$1,428.57$$
$$PV = \frac{PV_{11}}{(1+r)^{11}} = \frac{1,428.57}{(1+7\%)^{11}} = \$678.70$$

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV
- b. The bank's payments to you are a perpetuity deferred for 11 years

$$PV_{11} = \frac{C}{r}$$
$$PV_{11} = \frac{100}{7\%} = \$1,428.57$$
$$PV = \frac{PV_{11}}{(1+r)^{11}} = \frac{1,428.57}{(1+7\%)^{11}} = \$678.70$$

- c. Not a good deal (you pay \$749.83 for something worth \$678.70)

Growing Annuities

- Present value of a growing annuity, where g is the growth rate of the cash flows,

Growing Annuities

- Present value of a growing annuity, where g is the growth rate of the cash flows,

$$\text{PV of growing annuity} = \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right]$$

Take the Lump Sum or Growing Annuity?

- The Robinsons of Munford, Tennessee were one of the three winners in the “1.6 billion” Powerball jackpot in 2016
- They could choose between,
 - A. A lump sum of \$327 million, or
 - B. 30 installments of a growing annuity due with $g = 4\%$, starting at \$9.43m, with 30 payments, totaling around \$529 million



Photo, AP/Mark Humphrey

Take the Lump Sum or Growing Annuity?

- The Robinsons of Munford, Tennessee were one of the three winners in the “1.6 billion” Powerball jackpot in 2016
- They could choose between,
 - A. A lump sum of \$327 million, or
 - B. 30 installments of a growing annuity due with $g = 4\%$, starting at \$9.43m, with 30 payments, totaling around \$529 million
- Which option should they take?



Photo, AP/Mark Humphrey

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends
- Suppose the discount rate is 3%

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$
$$= \$326.58\text{m}$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

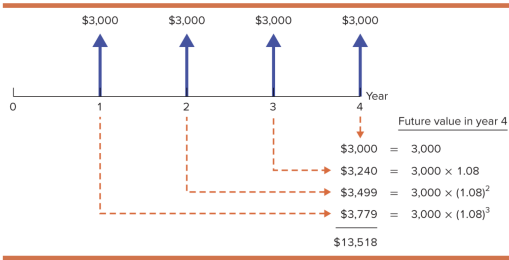
$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$
$$= \$326.58\text{m}$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

- That is, the present value of the annuity is almost exactly the same as (but slightly lower than) the \$327m lump sum

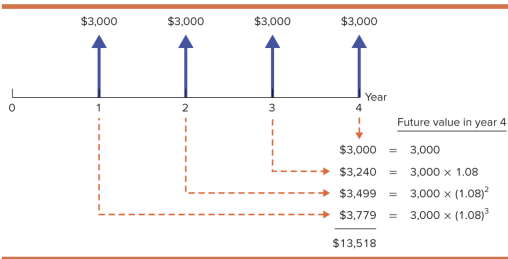
Future Value of Annuities

FIGURE 5.11 Calculating the future value of an ordinary annuity of \$3,000 a year for four years (interest rate = 8%).



Future Value of Annuities

FIGURE 5.11 Calculating the future value of an ordinary annuity of \$3,000 a year for four years (interest rate = 8%).



$$\text{FV of annuity of \$1 a year} = \text{PV of annuity of \$1 a year} \times (1 + r)^N$$

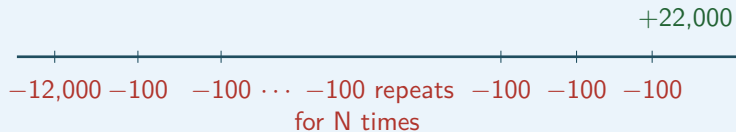
$$= \left[\frac{1}{r} - \frac{1}{r(1 + r)^N} \right] \times (1 + r)^N = \frac{(1 + r)^N - 1}{r}$$

Practice Problem

- I now have \$12,000 in the bank earning interest of 0.50% per month. I need \$22,000 to make a down payment on a house. I can save an additional \$100 per month. How long will it take me to accumulate the \$22,000?

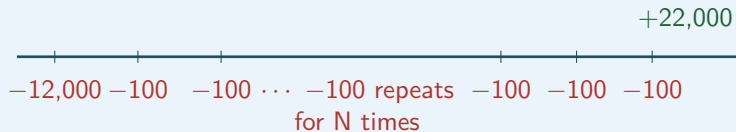
Practice Problem

- From your perspective, the cash flow is



Practice Problem

- From your perspective, the cash flow is



- You need to solve, future value (FV) of the \$12,000 + FV of the N \$100s = FV of the \$22,000

$$12000 \cdot (1 + 0.005)^N + 100 \cdot \frac{(1 + 0.005)^N - 1}{0.005} = 22000$$

Practice Problem

- The problem can, alternatively, be solved as, present value (PV) of the \$12,000 + PV of the N \$100s = PV of the \$22,000

Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

Practice Problem

- The problem can, alternatively, be solved as, present value (PV) of the \$12,000 + PV of the N \$100s = PV of the \$22,000

Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

- Let's recall that financial calculators are programmed to solve the following,

$$PV + \left[\frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

Practice Problem

- The problem can, alternatively, be solved as, present value (PV) of the \$12,000 + PV of the N \$100s = PV of the \$22,000

Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

- Let's recall that financial calculators are programmed to solve the following,

$$PV + \left[\frac{PMT}{(1+I)^1} + \frac{PMT}{(1+I)^2} \cdots + \frac{PMT}{(1+I)^N} \right] + \frac{FV}{(1+I)^N} = 0$$

Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

Practice Problem

- A couple will retire in 50 years. They plan to spend about \$20,000 a year in retirement, which should last about 25 years. They believe that they can earn a real interest rate of 9% on retirement savings
 - a. If they make annual payments into a savings plan, how much will they need to save each year? Assume the first payment comes in 1 year
 - b. How would the answer to part (a) change if the couple also realize that in 20 years they will need to spend \$50,000 on their child's college education?

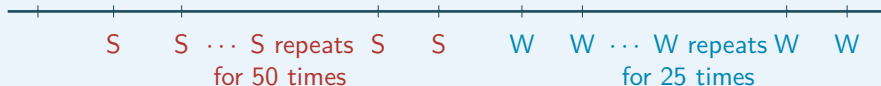
Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



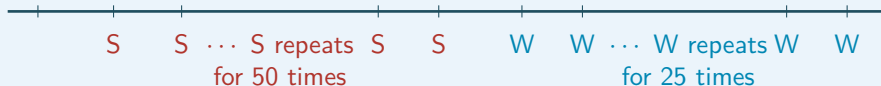
Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



- How much savings is required at the time of retirement?

$$PV_{\text{withdrawals, year 50}} = 20,000 \frac{1}{0.09} \left(1 - \frac{1}{(1 + 0.09)^{25}} \right) = 196,451.59$$

- On a calculator, $PMT = 20,000$, $I/YR = 9$, $N = 25$, $FV = 0$

Practice Problem

- The FV of the 50 savings, each of the amount $\$S$, should be that

Practice Problem

- The FV of the 50 savings, each of the amount \$S, should be that

$$FV_{\text{savings, year 50}} = S \left(\frac{(1 + 0.09)^{50} - 1}{0.09} \right) = 196,451.59 \Rightarrow S = \$241.02$$

- On a calculator, $FV = 196,451.59$, $I/YR = 9$, $N = 50$, $PV = 0$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, $PV \text{ of all savings, discounted to today} = PV \text{ of all withdrawals, discounted to today}$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, PV of all savings, discounted to today = PV of all withdrawals, discounted to today
- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, PV of all savings, discounted to today = PV of all withdrawals, discounted to today
- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

- Now determine the annual savings for 50 years that equates to that present value

$$\$11,563.53 = C \times \frac{1 - \frac{1}{(1+9\%)^{50}}}{9\%} \Rightarrow C = \$1,054.90$$

- On a calculator, $PV = 11,563.53$, $I/YR = 9$, $N = 50$, $FV = 0$

Practice Problem

- Professor's Annuity Corporation offers a lifetime annuity to retiring professors. For a payment of \$79,000 at age 65, the firm will pay the retiring professor \$575 a month until his death
- a. If the professor's remaining life expectancy is 20 years, what is the monthly interest rate on this annuity?
- b. If the monthly interest rate is 1.00%, what monthly annuity payment can the firm offer to the retiring professor?

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $PMT = 575$, $N = 240$, $PV = -79,000$, $FV = 0$. Find I/YR

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $\text{PMT} = 575$, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$. Find I/YR
- b.

$$C \times \left[\frac{1}{0.01} - \frac{1}{0.01 \times (1.01)^{240}} \right] = \$79,000 \Rightarrow C = \text{PMT} = \$869.86$$

- On a calculator, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$, $\text{I/YR} = 1$. Find PMT

Recap of Part 3

- A perpetuity is a level stream of cash flows at regular intervals that never ends

Recap of Part 3

- A perpetuity is a level stream of cash flows at regular intervals that never ends
- An annuity is a level stream of cash flows at regular intervals that ends after N periods (usually years, but the periods do not have to be years)

Recap of Part 3

- A perpetuity is a level stream of cash flows at regular intervals that never ends
- An annuity is a level stream of cash flows at regular intervals that ends after N periods (usually years, but the periods do not have to be years)
- These have handy formulas for calculating their present values!

Recap of Part 3

- A perpetuity is a level stream of cash flows at regular intervals that never ends
- An annuity is a level stream of cash flows at regular intervals that ends after N periods (usually years, but the periods do not have to be years)
- These have handy formulas for calculating their present values!
- Timing,
 - Ordinary perpetuities and annuities make their first payment at time $t = 1$
 - Perpetuity and annuity dues make their first payment at $t = 0$
 - Delayed perpetuities and annuities make their first payment at $t > 1$

Module 4, Part 4

Inflation

History of Inflation



Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.
Source: U.S. Bureau of Labor Statistics.

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
- **Nominal interest or nominal return**, the rate at which money invested grows
 - Compare dollars at time $t = 0$ to dollars at time $t = 1$

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
- **Nominal interest or nominal return**, the rate at which money invested grows
 - Compare dollars at time $t = 0$ to dollars at time $t = 1$
- **Real interest or real return**, the rate at which purchasing power increases
 - Compare the ability to buy a basket of goods at time $t = 0$ to the ability to buy the same basket of goods at time $t = 1$

Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

Approximation, $\text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$

Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

Approximation, $\text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$

- Inflation also affects discounting,
 - We should discount nominal dollar cash flows using a nominal discount rate
 - We should discount real dollar cash flows using a real discount rate

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?
- Answer,

$$1 + \text{Real interest rate} = \frac{1 + \text{Nominal interest}}{1 + \text{Inflation}} = \frac{1.06}{1.02} = 1.039$$

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?
- Answer,

$$1 + \text{Real interest rate} = \frac{1 + \text{Nominal interest}}{1 + \text{Inflation}} = \frac{1.06}{1.02} = 1.039$$

$$\text{Real interest rate} = 0.039 \text{ or } 3.9\%$$

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?

- Answer,

$$1 + \text{Real interest rate} = \frac{1 + \text{Nominal interest}}{1 + \text{Inflation}} = \frac{1.06}{1.02} = 1.039$$

$$\text{Real interest rate} = 0.039 \text{ or } 3.9\%$$

- Approximation, $\text{real} \approx 6\% - 2\% = 4.0\%$

Key Takeaways

- A dollar today is worth more than a dollar tomorrow, $FV = PV \times (1 + r)^T$ and $PV = FV / (1 + r)^T$ translate between them

Key Takeaways

- A dollar today is worth more than a dollar tomorrow, $FV = PV \times (1 + r)^T$ and $PV = FV / (1 + r)^T$ translate between them
- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment

Key Takeaways

- A dollar today is worth more than a dollar tomorrow, $FV = PV \times (1 + r)^T$ and $PV = FV / (1 + r)^T$ translate between them
- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment
- The real return strips inflation out of the nominal return

Thank you!

Please let me know if you have any questions.